1. Introduction and executive summary

For a large logistics company in India, three challenges are obvious: late deliveries, expensive fuel and uncertain demand. We describe how it addresses these challenges using AI - by optimizing routes, providing accurate demand prediction, and automating its warehouse operations. These steps lead to enhanced efficiency and customer satisfaction. And the three benchmark companies: Delhivery (India); Amazon Logistics (Global); and Maersk (Global).

The main point is that the three challenges are actually the same challenge from 3 different perspectives. We do not plan well, anticipate well, or execute well. The solution is to have a connected four-layer AI stack (Fig. 4) where each layer informs the next. To summarize in one sentence: the company uses supervised regression and deep sequence models to predict demand; uses this forecast to place stock; uses A* and metaheuristics at scale of the fleet to plan routes; uses A*, Reinforcement Learning, Computer Vision, Agentic AI to run its warehouses; uses humans-in-the-loop and explainable models. We have attached all the supporting tables and graphs in the Appendix (Tables A1 - A15, Figs. 1-5). This way, the main content focuses on the main argument.

1.1 Scope, method and how the two drafts were combined

We cover all pieces of parcel network: 1st mile pickup, linehaul between hubs, hub+warehouse operations, last mile delivery. Scope of solution is only about decisions we can make better with data we have or can get (orders, gps & telematics, traffic, weather feeds, warehouse scans, fuel logs) with AI and no new vehicles/hubs. Approach is: 1. go from symptoms to root causes, 2. spot what question you are asking (find path, predict number, group items, learn from feedback) - as question determines technique, 3. link each technique to its slide deck source 4. link each module to KPI.

Two drafts we had were complementary. 1st one (case report) gave argument, example, risk register, decision gates. The 2nd one (mapping note) more rich technical (NLP address geocoding, Iot & Edge Ai, predictive maintenance, multi-agent robots), benchmarked companies, Unit 5 implementation toolkit and stakeholder view. On disagreements this report retains more cautious one. Unsourced numbers only as labeled planning targets (Section 8) and 18mo phase plan sits inside 3 horizon (Section 10).

2. Problem analysis

Every symptom has to be linked back to a root cause before a method can be selected (Table A1). Delays in delivery are due to static routes that don't take into account real time traffic, weather and failed attempts, plus poor arrival time estimates. High fuel cost is due to extra km, idling, poor stop sequencing and part loaded or empty backhaul. Unpredictable demand is a result of forecasts based on averages which don't take into account festivals, monsoon and promotions and so stock ends up in the wrong hub. Inefficiency in the warehouse is due to long manual walks, slow moving stock being located close to dispatch and picking errors. Poor customer satisfaction is the consequence of late or failed deliveries and the lack of any proactive updates.

The mapping note has a point about why the old ways of doing it simply don't work. The static route plans and the spreadsheet forecasts are all rule based (IF-THEN) and can't deal with the uncertainty, the conflicting information or the rapidly changing variables that you're likely to come across in the real world (monsoon closures, an Indian address based on a landmark etc). That's the course's explanation of why symbolic AI failed and why learning based AI (which learns as new data comes in) is the correct foundation.

2.1 How the problems feed each other

The symptoms link together. A bad demand forecast means stock ends up in the wrong place, leading to emergency transfers and last-minute vehicle hires—fuel costs go up. Static routes and slow warehouses push dispatch back, causing parcels to leave the hub late and delays pile up on the road. Late or failed deliveries lead to complaints and repeat attempts, adding more kilometres and fuel. Tackling just one symptom won’t do much: improving routing cannot make up for lost time in a slow warehouse, and a faster warehouse doesn’t help if the wrong stock is there. This is why the company creates one connected system.

2.2 What success looks like

What does success look like? It’s defined by direction and measurement, not just promises. On-time delivery and first-attempt success improve; kilometres and fuel spent per parcel decrease; forecast error drops below the current method, with fewer stock-outs and overstock days; picks per hour increase and picking errors decrease; customers get early, accurate updates. Each of these is monitored against a baseline period or control group during the pilots (see Table A8).

3. Proposed solution: a four-layer AI stack

The solution ? A pipeline—not four separate tools. It's the "AI Demand Sensing, Inventory, Digital Twin and Replenishment" scenario from the course auction slides, but now it’s got routing and warehouse execution added (see Table A2, Figure 4). This setup includes all four essential elements of an AI system (Unit 1): Perception (cameras, barcode scanners, vehicle IoT), Understanding and reasoning (parsing addresses, diagnosing equipment wear), Learning (models trained on historical orders and delivery pings), and Decision and action (rerouting, sorting, replenishment). The routing and forecasting engines? Those are Limited Memory AI: they remember past traffic, delivery times, and festival sales to make better decisions later.

- Data foundation. Orders, GPS, telematics, traffic, weather, festival calendars, warehouse scans, fuel logs—all of it goes into one neat store. Data fuels AI, and low quality? That's expensive (S02, S03). Structured data (ERP, WMS, TMS tables, SKU weights, pin codes) joins with unstructured data (free-text addresses, camera images, social signals) using semantic tech and knowledge graphs, making it machine-understandable rather than just machine-readable.
- Predict. Regression, regression trees, ensembles, LSTM/RNN models, and probabilistic reasoning—they forecast orders per hub per day, estimate delivery times, and raise spike alerts, always as ranges.
- Plan. The forecast sets stock targets and capacity per hub. A*, greedy search, and genetic, annealing, tabu or ant-colony methods? They build routes within vehicle and rider limits.
- Execute. A*, reinforcement learning, computer vision, multi-agent robots, and agentic AI manage pick paths, quality checks, and automatic replenishment.

A feedback loop sends back actual delivery times, stock-outs, and traffic results to the Predict layer: that’s the Learning part of AI from S01. Five design principles steer the build: start simple and clear (baselines and trees before diving into deep models), involve humans for major or strange decisions, pilot in one city and one hub, retrain weekly based on outcomes, and evaluate every module by business results—not just model accuracy.

3.1 The layers in more detail

Data Foundation. The system carries out data quality checks by validating addresses. It also picks up the gaps in GPS data. The system also detects the situation of scanning the same item twice. It also detects the multiple usage of the same hub code. The respective hub is to resolve data issues at the time of data entry. Predict. This system predicts duration for the delivery process using demand forecasting. This process involves studying orders for different days of the week. This system also considerss various factors like public holidays, weather and local community happening. The sub-system that carries out this process generates numbers that are within limits. These numbers help the next sub system to know where there may be issues or unclear information. Plan. It shows the required supplies and needed numbers of riders, vehicles and shifts based on the forecast. Routing tools used to generate paths with set rules. Also calculation of cost, such as fuel usage. Execute. Robots and workers will follow the possible paths. All packages are monitored by camera. Agent will make requests for stock. Within the allowed limits. Each time when the deliveryers start working. They are provided with directions.

4. Module 1: Route optimisation (delays and fuel)

4.1 Formulating the problem and choosing the algorithm

State, initial state, goal, action, transition model and path cost are all required in a search problem. State is current location + parcels already delivered for a rider. Goal is delivery of all parcels. Action is to drive to next stop. Path cost is a weighted combination of travel time in live traffic + distance + fuel. So attacking fuel directly. All stops + truck capacities + roads are the state space we search over with the algorithms.

There's no single algorithm that works in all cases (see A3 table). A* search to find best route between 2 points in live traffic sorts the options by f(n) = g(n) + h(n) (cost so far + estimated cost to go). Optimal if never overestimates. (recommended way to do delivery/ridehailing paths in the course). Greedy best first. Good for instant re-route when you're having a sale at a festival or flood. Quickness takes precedence over perfect. There's explosion in the number of combinations for 50-200+ stops per vehicle. So they use metaheuristics. Simulated annealing for vehicle routing problem (Delhivery, Shadowfax example in slides). Genetic for travelling salesman style ordering of thousands of orders (e.g. Flipkart, Amazon). Ant colony optimisation for dynamic last mile routing. Tabu for network level hub design. K-meansing of drop points into tight groups for rider. Supervised regression for arrival time prediction. Reinforcement learning (e.g. Deep Q-nets) to adapt to changing roads through reward for on-time, penalise late/heavy fuel runs. Beware they do BFS in the slides for a "12 stops" example because the map is unweighted. In reality we have unequal travel times on roads and you should do a weighted method like A* or uniform-cost in prod.

4.2 Worked example: why A* beats greedy

Rider going from hub (H) to customer cluster (C), via stop A or B (Table A4, Figs 1 and 2). Estimate for A being close to C is 5, but road is congested, actual remaining cost is 40. Estimate for B being far is 10, actual cost is 12. Greedy best-first uses only estimates. Chose A and ended in 10+40=50 minutes. A* expands A first (f=15). Cost to continue is 50 total. Then expands B (f=25). Gets to C in 15+12=27 minutes. Both estimates are admissible. A* guaranteed to give best route. Not necessarily the cheapest next move to do is on the cheapest path. Just for illustration.

4.3 Fuel in the cost function and fleet-scale planning

Cost function: cost = w_time*time + w_distance*distance + w_fuel*fuel (where ws are set by management) estimate fuel used based on type of vehicle load speed idling etc. so you can choose longer roads less stopping/going can use a* heuristic and still be admissible by changing line distance to min time or fuel can switch to lower fuel mode for premium parcels or slower lower fuel for standard parcels by changing weights

For fleets do in stages k-means to get rider zones then order stops in zone using ant-colony method (with crossover mutation or pheromone reinforcement) then find best plan using simulated anneal (or hill climb + restart) (use annealing so can occasionally make worse moves to escape local optima) do a* for each leg

Make these costs part of problem.

(ie. include vehicle capacity, time length of shifts, time windows, and backhaul match (so reduce return trips))

CVRPTW - capacitated vehicle routing problem with time windows. can use default reason to plan assuming the vehicles on time and then update when get a delay reported

4.4 Address intelligence, telematics and customer updates

A lot of missed deliveries start with addresses. As far as India goes, it would typically be free-form addresses such as "near old banyan tree, behind the bank". That means, as you process the address, you use an NLP with a transformer architecture. The encoder processes text. Positional encoding preserves the word sequence. Self and multi-headed attention assigns importance to critical words such as city, locality, landmark to map into coordinates. (The reference implementation is the NetMap address engine in Delhivery.) From the vehicle side, you stream speed, idle, fuel data from OBDII and IoT sensors. Part of your data preprocessing in the truck is edge AI. It handles constraints around computation and connectivity capabilities. It delivers eco-driving coaching, too. For prediction of time of arrival, you run a supervised regression algorithm with past trips and traffic/weather/remaining stops as features. For identification of parcels that may be delivered with delay, you run a classifier. The algorithm warns customers before the delivery attempt. You preconfirm address and available times before the actual delivery.

4.5 Measuring routing results

The routing pilot compares AI-planned routes with control routes in the same city over the same period. The company tracks on-time delivery, OTIF, first-attempt success, kilometres and fuel per parcel and cost per delivery (Table A8). Because festival weeks distort comparisons, the pilot runs long enough to include at least one demand peak, and results are reported as ranges. They guide the weights in the cost function and the decision to extend routing to more cities.

5. Module 2: Demand prediction

5.1 Choosing the learning type and the model

Decisive test that ask what the output is. Demand is a number (orders per hub, pin code and day), so this is a regression task under supervised learning trained on historical input and output pairs (Table A5). The company starts with a moving average baseline of the regression tree because planners can read its rules , then ensembles such as random forests which combine many trees to raise accuracy and manage overfitting and the bias variance and trade off. A recurrent network (RNN or LSTM), and later transformer models, capture weekly cycles and festival ramps ; feature engineering, representation learning and backpropagation are the underlying course concepts. The best performer by business outcome is kept.

Unsupervised methods add value around the forecast that shows K-means or DBSCAN segment pin codes by demand behaviour, and Isolation Forest detects sudden spikes or data faults so planners can tell a real surge from a broken feed. Probabilistic reasoning (Bayes' theorem, confidence ranges) states uncertainty, and fuzzy logic turns vague signals such as busy festival week into a degree between 0 and 1 .

5.2 Features, evaluation and the five-module architecture

Internal features are past orders by pin code and day, lag values (last week, same day last year), rolling averages, prepaid versus cash-on-delivery mix and return rates. External signals are festivals and sale events, monsoon weather, local

events, competitor promotions and road closures. Structural features are hub capacity, pin-code type (metro, Tier-2, rural) and seasonality flags. Data is split by time, training on earlier months and testing on later ones, never at random, or the model peeks at the future. Quality is judged by MAPE and RMSE and by outcomes: stock-outs, overstock days and last-minute vehicle hiring. The output is a range, not a single number, so safety stock sits where uncertainty is highest.

This pillar contains the five modules required by Scenario 2 of the course auction: an AI demand sensing platform for SKU and pin-code demand; an external signal integration module for weather, social and competitor data; a multi-echelon inventory optimisation engine that pre-positions stock across regional hubs and dark stores; a digital twin of the supply network that simulates festive spikes and the bullwhip effect; and an automated replenishment decision system. Models are retrained weekly and monitored daily for drift, for example after a change in customer mix.

5.3 From forecast to decision

Consider a forecast that shows a surge at one hub on a festival weekend. The company moves stock to that hub beforehand, reserves vehicles and riders, pre-assigns delivery zones and schedules an extra warehouse shift. Without the forecast each step happens late and at a premium, through emergency transfers and spot hiring. The forecast therefore sits upstream of routing and warehouse work.

5.4 Model workflow, uncertainty and special events

The model is built in a fixed sequence: clean the data, create features (lags, rolling averages, event flags), split by time, set the moving-average baseline, fit trees and ensembles, test an LSTM if enough history exists, compare candidates on MAPE, RMSE and business outcomes, deploy overnight forecasts per hub and pin code, monitor errors daily and retrain weekly or whenever accuracy drifts. Demand is never known exactly, so the company reasons with probabilities: Bayes' theorem updates the belief about tomorrow's volume as early signals arrive, such as first-hour orders on a sale day , and Isolation Forest separates a real surge from a broken data feed. For known peaks the digital twin runs what-if scenarios before the event, and safety stock is held where the forecast range is widest.

6. Module 3: Warehouse automation and smart fulfilment

The warehouse module matches each task to a technique (Table A6). Slotting comes first: forecast velocity per SKU, cluster SKUs into fast, medium and slow movers and place fast sellers near dispatch, which shortens every walk before any robot is involved. Association rules (market-basket analysis, an unsupervised method) place frequently co-ordered items together, and simulated annealing is the slide-deck method for warehouse layout, deciding product placement to cut picking time, as at Flipkart and Amazon.

Path planning: aisles and racks form a graph, and A* finds the shortest path between pick locations, re-planning when an aisle is blocked ; ant colony optimisation supports collision-free robot paths. The order in which pick locations are visited is a combinatorial problem like road routing, so a genetic algorithm or local search sets the sequence and A* handles each leg; this pairing is a design extension beyond the slides. Learning robots: reinforcement learning gives a reward for fast, collision-free picks and a penalty for errors, as in the "robotic warehouse picker" , trained first in the digital twin so mistakes do not damage stock. In agent terms, a pick robot is model-based at the base because it holds a warehouse map, goal-based when it plans to a target, and a learning agent once reinforcement learning improves its policy.

Vision and sorting: overhead cameras are the Perception layer, and convolutional neural networks use sliding filters to detect damaged parcels, wrong or unreadable labels and empty shelves in milliseconds. Uncertain items go to a human checker, and their corrections become labelled training data. Rules and agents: a transparent rule-based system handles stable compliance logic (heavy parcels to the heavy lane, incomplete addresses held). An agentic replenishment system, given a goal such as keeping stock-outs below target, reads the forecast and stock, then raises purchase or transfer orders: small orders inside limits are automatic, larger ones need manager approval, and every action is logged. Fleets of autonomous mobile robots (AMRs) work as a multi-agent system under goods-to-person logic, where robots bring racks to people, which is human-AI collaboration and not replacement. Predictive maintenance follows the Scenario 6 blueprint: an industrial IoT sensor network, Edge AI devices, predictive models, a digital twin of critical machines and automated work orders. Vibration, acoustic and thermal data from conveyors and sorters are monitored, and autoencoders flag anomalies before breakdowns.

6.1 Guardrails and measurement

Warehouse gains are measured in picks per hour, order cycle time, walking distance per order, picking error rate and equipment downtime, against the same hub's baseline. Five guardrails protect operations: human checkers review uncertain vision results, agent orders above set limits need approval, a rule-based layer enforces weight limits and

address checks, new behaviour is trained and tested in the digital twin before it goes live, and every automated action is logged so it can be audited.

7. Integration and benchmarks from the three companies

A digital twin of the supply network is a live simulation of hubs, vehicles, stock and demand. Planners test "what if a flood closes this highway" or "what if Diwali demand rises 30%" before it happens, and it is the safe place to train reinforcement learning agents. The daily flow (Table A7) runs from overnight forecasts to stock moves, zone assignment, picking, live rerouting and weekly retraining.

7.1 Customer communication and continuous learning

The stack ends with the customer. Predicted delays, route changes and failed-attempt risks become proactive messages, so customers hear about a delay from the company before they notice it, and their replies and complaints are logged as feedback. Every day also produces outcomes (actual delivery times, stock-outs, traffic, warehouse errors) that flow back into the data foundation and retrain the models weekly. This closed loop is the Learning element of AI from S01 and keeps the system accurate as the network and the market change.

7.2 Lessons from Delhivery, Amazon Logistics and Maersk

The three assigned companies show the same ideas at scale (Table A12). Delhivery is the Indian benchmark: its NetMap address engine shows NLP and knowledge representation overcoming unstructured data, and its dynamic mesh routing reflects simulated annealing for the vehicle routing problem and agentic decision-making. Amazon Logistics appears across the slides for A* in warehouse picking, simulated annealing for layout, genetic algorithms for delivery routing, reinforcement learning for robotic pickers and goal-directed agents. Maersk shows digital twins, A*-style heuristic search that balances fuel already used against estimated remaining fuel under weather and current constraints, green AI that cuts marine bunker fuel emissions, and Edge AI predictive maintenance on refrigerated containers. The lesson for the Indian company is that no benchmark relies on one clever model; each connects data, planning and execution.

8. Business value, KPIs and ROI

Every module is tied to a measurable result (Table A8): on-time delivery and fuel per parcel for routing, forecast error and stock-out rate for demand, picks per hour and error rate for the warehouse, and NPS or CSAT for customers. The ROI logic is simple. Annual benefit is fuel savings plus savings from fewer failed deliveries, labour and overtime savings, inventory holding savings and recovered sales. Annual cost is cloud and data infrastructure, data and ML engineering, warehouse hardware, and change management and training. ROI = (benefit - cost) / cost. Routing and forecasting need mainly software and data, so they can show returns earlier than the hardware-heavy warehouse module, which is why they are piloted first.

8.1 Planning targets and how they will be tested

The curriculum-mapping draft stated planning targets of 22% to 30% higher on-time-in-full (OTIF) delivery, 15% to 20% lower fuel cost (12% to 18% less fuel burn from eco-driving and routing) and about 20% lower safety-stock holding cost (Figure 5). No source for these figures is given, so this report treats them only as planning assumptions to be proved or rejected by pilots against a baseline or control group, with results reported as ranges. Savings that cannot be traced to a source are not quoted.

8.2 Measuring success fairly and benefits beyond cost

Every pilot is compared with a baseline period or control group so that gains are not confused with seasonal changes, the same KPIs are recorded before and after, and the same city and period are used for control routes. This discipline protects the programme, because leaders can trust the numbers when they decide whether to scale. Some benefits are harder to price but matter: customers gain reliable delivery windows and early warnings, which supports retention; planners gain forecasts with ranges and explanations; and the network becomes more resilient because disruptions such as floods or sale-day surges are rehearsed in the digital twin.

9. Risks, ethics and change management

AI ethics and the NovaBanc case make fairness, transparency and human oversight part of every solution (Table A9).

9.1 Data and model risks

Wrong addresses, missing GPS points and gaps in scans corrupt both forecasts and routes. The company counters this with data cleaning, address validation and hub-level ownership of data quality. Models also drift as customers, products and roads change, so accuracy is monitored daily and models are retrained weekly. To avoid black-box decisions, it

starts with explainable models (regression trees and rules) and adds dashboards that show why a route or forecast was chosen, because dispatchers and managers use a system more readily when they can see its reasoning.

9.2 Fairness and regional coverage

The main ethical risk is bias toward profitable areas. As in the NovaBanc loan case, a model trained on past decisions can repeat past bias: if history shows lower profit in Tier-2, Tier-3 and rural pin codes, it may learn to serve them last. The company reports service levels by region, sets minimum service floors for every pin-code type and audits results regularly. If a region falls below its floor, the issue is escalated to a human decision maker and is not left to the model.

9.3 People, privacy and over-automation

People risk decides whether the system is used. The Unit 5 toolkit identifies five stakeholder groups (Table A13): the board wants ROI and risk control, the CFO payback and budget control, the CTO technical feasibility and integration, middle managers relevance and workload, and frontline staff job security and ease of use. The company explains that work shifts from manual planning and long walks to supervision and exception handling, offers reskilling, shares gains such as fewer overtime hours and frames robots and eco-driving coaching as augmented intelligence. The "Drishti Diwali Disaster" case is the warning: AI deployed as a top-down, one-shot IT project during a festive peak fails, so the remedy is a change-managed, human-in-the-loop transformation with hub teams involved in the pilots. Rider tracking and customer addresses are personal data, so India's Digital Personal Data Protection Act, 2023 applies: the company limits collection, obtains consent where required and restricts access by role. For agentic systems it sets approval thresholds for large actions, logs every action, monitors behaviour and keeps a kill switch so humans can stop the agent at once.

10. Implementation roadmap, readiness and conclusion

The programme runs about 18 months in five phases (Table A10, Figure 3): Phase 0 (months 0 to 3) builds the data foundation, Phase 1 (months 3 to 6) pilots routing in one or two cities, Phase 2 (months 4 to 8) adds demand forecasting, Phase 3 (months 6 to 10) pilots warehouse automation in one hub, and Phase 4 (months 10 to 18) scales to all hubs and adds the digital twin and automatic replenishment.

10.1 Phase activities and exit criteria

In Phase 0 the company unifies order, GPS, traffic, weather and warehouse data, fixes quality gaps and agrees data ownership at each pilot hub; the exit criterion is a clean, accessible data set. In Phase 1 A*, fleet optimisation and address NLP run alongside control routes; the exit criterion is lower kilometres and fuel per parcel and a better on-time rate. In Phase 2 hub and pin-code forecasts with external signals are compared with the current method; the exit criterion is lower forecast error and fewer stock-outs. In Phase 3 path optimisation, slotting and vision checks are tested in one hub; the exit criterion is higher picks per hour and fewer picking errors. In Phase 4 the solution is rolled out to every hub with simulation and automatic replenishment; the exit criterion is network-wide KPI gains and a positive ROI.

10.2 Governance, team and decision gates

A steering group of operations, technology and finance leaders owns the programme and reviews results at each gate. A central data and machine-learning team builds and maintains the models, and each hub names an operations champion who tests outputs against real conditions and gathers feedback from riders and warehouse staff, which keeps the models connected to daily work. At the end of each pilot the group compares KPIs with the baseline or control group. If the criteria are met the module moves to the next phase; if not, the team adjusts the model, the data or the process and repeats the pilot. Investment therefore stays tied to proven results, and nothing is scaled before it has worked.

10.3 Horizons and readiness

This plan fits the three-horizon view. Horizon 1, quick wins (0 to 6 months) covers the data pipelines, IoT telematics, address geocoding and the first routing pilot, building trust through visible fuel and delivery wins. Horizon 2, core transformation (6 to 18 months) scales forecasting, CVRPTW routing and CNN sorting across hubs. Horizon 3, strategic differentiation (18 to 36 months) connects AMRs, digital twins and agentic AI into a closed-loop capability competitors cannot easily copy. The 3x3 readiness matrix explains why Phase 0 comes first: data readiness (clean ERP, WMS and TMS pipelines), talent readiness (ML Ops, data and AI engineers) and process readiness (documented sorting and dispatch workflows). The slides cite research that most AI projects fail to scale because the organisation was not data-ready.

10.4 Priority recommendations

- Start with data and routing. Phase 0 and the routing pilot give the fastest visible gains in fuel and on-time delivery and build trust for later phases.

- Match method to question. Use fuel-weighted A* for point-to-point paths, metaheuristics for whole fleets and weighted search instead of breadth-first on real roads.
- Climb the model ladder. Beat a moving-average baseline with a regression tree before investing in LSTM or transformer forecasts.
- Build fairness and explainability in from day one. Service floors by pin-code type, explanation dashboards and approval thresholds cost little early and a great deal to retrofit.
- Treat targets as hypotheses. Quote savings only after pilots show them against a baseline or control group.

Assumptions and limitations. The A* example uses illustrative numbers, and the KPI targets in Figure 5 are planning assumptions without a cited source. Mappings to course slides and to Delhivery, Amazon Logistics and Maersk follow the course materials and the two source drafts; they are not an independent audit of those companies' systems. Results will depend on the quality of the company's own data, the length of the pilots and the willingness of hub teams to use the tools, which is why the roadmap measures every step against a baseline before scaling.

Outcome by problem. Delayed deliveries are addressed by continuous re-planning with A*, arrival-time prediction and early customer notification. High fuel cost is addressed by fuel in the path cost, compact rider zones, backhaul matching and eco-driving coaching. Unpredictable demand is addressed by forecast ranges built on external signals and by pre-positioned stock. Warehouse inefficiency is addressed by slotting, optimised pick paths, vision checks and agentic replenishment, and customer satisfaction follows from reliable delivery and proactive updates.

Conclusion. Delays, fuel cost and demand swings share one cause: planning, prediction and execution are disconnected. A four-layer AI stack that forecasts demand, positions stock, plans routes with A* and fleet optimisation, and automates the warehouse addresses all three together. Method choice follows the course test of identifying the question first (Table A11): a path with a cost estimate calls for A*, a number for regression, groups for clustering and trial-and-error learning for reinforcement learning. Success depends on clean data, explainable models, fairness across regions and a pilot-first rollout that proves value with real numbers.

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Delhivery AI Case Study | How AI Optimizes Logistics and Last-Mile Delivery

How can generative AI drive logistics transformation │ Maersk

How is Maersk using AI? (2026)

Appendix: Tables and Figures

Tables A1 to A15 and Figures 1 to 5 support the main text. Figures 1 and 2 use the illustrative numbers in Table A4; Figure 3 uses the roadmap in Table A10; Figure 5 plots planning targets, not measured results.

Tables

Table A1. Root-cause map

Symptom	Likely root cause	What AI must do	Course unit
Delayed deliveries	Static routes; traffic, weather and failed attempts not handled in real time; poor ETA estimates	Re-plan routes continuously and predict ETA	Unit 2 (search), Unit 4 (supervised)
High fuel cost	Extra kilometres and idling; poor load and stop sequencing; empty or part-loaded runs	Put fuel, time and distance into path cost; optimise the whole fleet plan; cut empty backhauls	Unit 2 (A*, local search, GA, ACO)
Unpredictable demand	Average-based forecasts; festivals, monsoon and promotions ignored; stock in the wrong hub	Forecast at pin-code and hub level; quantify uncertainty	Units 3 and 4
Warehouse inefficiency	Long manual walks; fast sellers far from dispatch; picking and sorting errors	Optimise picking, slotting and replenishment; automate checks	Units 1, 2 and 4
Weak customer satisfaction	Late or failed deliveries; no proactive updates	Predict delays and notify customers; learn from feedback	Units 4 and 5

Table A2. Four-layer AI stack

Layer	What it does	Inputs	Outputs	Core technique
1. Data foundation	Collects and cleans data from every source	Orders, GPS/IoT telematics, traffic, weather, festival calendar, scans, fuel logs	One clean data store	Data quality, semantic technologies, knowledge graphs (S02, S03)
2. Predict	Forecasts demand, ETA and spikes	History plus external signals	Orders per hub per day, ETA, spike alerts	Regression, trees, LSTM/RNN, Bayes
3. Plan	Positions inventory and plans routes	Forecasts, capacity, live traffic	Stock targets per hub, optimised routes	A*, greedy, GA/SA/ACO/Tabu, K-means
4. Execute	Runs warehouse and last mile	Orders, pick lists, route plans	Robot paths, pick sequences, auto replenishment	A*, RL, CNN vision, agentic and multi-agent AI

Table A3. Routing method selection

Situation	Method	Reason (course reference)
Best route between two points with live traffic	A* search, $f(n) = g(n) + h(n)$	Balances cost so far and estimated cost to go; optimal with an admissible heuristic (S05, S06)
Instant reroute during a festival sale or flood	Greedy best-first	Fast and good enough when speed matters (S05, S06)
Shortest path by number of stops, unweighted map	Breadth-first search	Guarantees fewest stops with a queue, but ignores travel time; not for real roads
Plan 50 to 200 stops for a vehicle or fleet (CVRPTW)	Genetic Algorithm, Simulated Annealing, Ant Colony, Tabu Search	Too many combinations for exact search (S07, S08)
Quick improvement of an existing plan	Hill climbing with restarts, or simulated annealing	Improves step by step; annealing escapes local optima (S07, S08)

Situation	Method	Reason (course reference)
Assigning parcels to riders by area	K-means clustering	Groups nearby drop points into compact zones (S13, S14)
Predicting arrival time	Supervised regression	Predicts a number from trips, traffic and weather (S13, S14)
Adapting to changing roads over time	Reinforcement learning (e.g. DQN)	Learns a policy from reward and penalty (S13, S14)
Cleaning free-text Indian addresses	NLP with Transformer encoder	Self-attention and positional encoding map landmarks to coordinates (Unit 4)

Table A4. Worked example: A* versus greedy (illustrative, minutes)

Node	Cost so far g(n)	Heuristic h(n)	f(n) = g + h	Real remaining cost	Total route via node
A	10	5	15	40	50
B	15	10	25	12	27

Greedy chooses A (smallest h) and takes 50 minutes. A chooses the B route and takes 27 minutes, a saving of 23 minutes.*

Table A5. Demand prediction methods

Task	Learning type	Method	Course reference
Forecast orders per hub, pin code and day	Supervised, regression	Moving-average baseline, regression tree, random forest, XGBoost/LightGBM	S09, S10; S13, S14
Time-ordered patterns (weekly cycles, festival ramps)	Supervised, deep learning	RNN, LSTM, transformer	S17, S18
Segment pin codes or SKUs by demand behaviour	Unsupervised	K-means, DBSCAN	S13, S14
Detect sudden spikes or data faults	Unsupervised (anomaly detection)	Isolation Forest	S13, S14
Express uncertainty	Probabilistic and inductive reasoning	Bayes' theorem, confidence ranges	S11, S12
Vague signals such as "busy festival week"	Fuzzy logic	Membership degrees 0 to 1	S11, S12

Table A6. Warehouse automation methods

Warehouse task	Method	How it works	Course reference
Shortest path for robots and pickers	A* on the warehouse grid; ACO for collision-free paths	Aisles and racks form a graph; A* avoids blocked aisles	S05, S06
Order of visiting pick locations	GA / local search / ACO	Combinatorial sequencing, then A* for each leg	S07, S08 (design extension)
Layout and slotting	Forecast, clustering, simulated annealing, association rules	Fast movers near dispatch; co-ordered items together	S07, S08; S13, S14
Robots that improve with experience	Reinforcement learning	Reward for fast, collision-free picks; penalty for errors	S13, S14
Damage, label and stock-out detection	Computer vision (CNN)	Cameras and image models spot damage, wrong labels, empty shelves	S01; S17, S18

Warehouse task	Method	How it works	Course reference
Stable compliance rules	Rule-based system	Transparent, auditable if-then rules	S07 to S10
Automatic replenishment and robot fleets	Agentic AI, multi-agent systems	Agent raises orders with human approval for large actions	S13, S14; Unit 5
Equipment breakdown prevention	Predictive maintenance, autoencoders	Sensor streams reveal anomalies in conveyors and sorters	Scenario 6; Unit 4

Table A7. Daily decision flow

Step	Action	AI used
1	Overnight, forecast tomorrow's orders per hub and flag spikes	Regression, LSTM/RNN, anomaly detection
2	Set stock targets and move inventory between hubs	Forecast-driven multi-echelon inventory optimisation
3	Assign parcels to rider zones and sequence stops	K-means, GA / ACO / simulated annealing
4	Generate pick lists and robot paths	A*, reinforcement learning, multi-agent coordination
5	Reroute around traffic and failed deliveries; notify customers	A*, greedy, ETA prediction
6	Log outcomes and retrain models weekly	Feedback loop

Table A8. KPIs and sources of value

Module	KPIs to track	Where the money comes from
Routing	On-time delivery %; OTIF; first-attempt success; km and fuel per parcel; cost per delivery	Lower fuel, fewer re-attempts, less overtime, better vehicle use
Demand prediction	Forecast error (MAPE, RMSE); stock-out rate; overstock days; last-minute vehicle hiring	Less lost sales, lower holding cost, planned instead of spot capacity
Warehouse	Picks per hour; order cycle time; picking error rate; walking distance per order; equipment downtime	Higher throughput per shift, fewer errors and returns
Customer	NPS / CSAT; complaints; return-to-origin rate	Retention and repeat orders, lower support cost

Table A9. Risks and mitigations

Risk	Why it matters	Mitigation
Poor data quality	Wrong addresses, missing GPS and scan gaps corrupt forecasts and routes	Data cleaning, address validation, hub-level data ownership
Bias toward profitable areas	Models may deprioritise Tier-2, Tier-3 and rural pin codes (NovaBanc pattern)	Service reporting by region, service floors, regular audits
Black-box decisions	Dispatchers will not trust unexplained routes or forecasts	Start with explainable models; explanation dashboards
Workforce resistance	Fear of job loss or surveillance	Reskilling, human approval loops, early communication, shared gains
Privacy	Rider tracking and addresses are personal data (DPDP Act, 2023)	Data minimisation, consent, role-based access
Over-automation	Agentic systems acting without limits can cause costly errors	Approval thresholds, action logs, kill switch
Failed one-shot rollout	Top-down deployment at a festive peak (Drishti Diwali Disaster)	Phased pilots, change management, human in the loop

Table A10. Implementation roadmap

Phase	Timeline	Scope	Exit criterion
0. Data foundation	Months 0 to 3	Unify order, GPS/IoT, traffic, weather and warehouse data; agree data ownership	Clean, accessible data set for pilot hubs
1. Routing pilot	Months 3 to 6	A*, fleet optimisation and address NLP in one or two cities	Lower km and fuel per parcel; better on-time rate than control routes
2. Demand forecasting	Months 4 to 8	Hub and pin-code forecasts with external signals	Forecast error below old method; fewer stock-outs
3. Warehouse pilot	Months 6 to 10	Path optimisation, slotting, vision checks in one hub	Higher picks per hour, fewer errors
4. Scale and digital twin	Months 10 to 18	Roll out to all hubs; simulation and automatic replenishment	Network-wide KPI gains and positive ROI
Horizon 3 (extension)	Months 18 to 36	Closed-loop autonomous supply chain: AMRs, twins, agentic AI	Sustained advantage; audited fairness and safety

Table A11. Method selection cheat sheet

If the question is...	Use	Because
Find the best path with a good estimate of remaining cost	A*	Optimal and efficient with an admissible heuristic
Decide quickly, near enough is fine	Greedy best-first	Fast but not optimal
Order many stops with huge combinations	GA / SA / ACO / Tabu	Handle large search spaces
Predict a number	Regression (supervised)	Output is numeric
Predict a category (late / on time)	Classification (supervised)	Output is a label
Find groups or odd cases without labels	Clustering / anomaly detection	No labelled answers needed
Learn the best action by trial and error	Reinforcement learning	Reward and penalty feedback
Reason with probabilities	Bayes, probabilistic reasoning	Handles uncertainty
Need an explainable decision	Rule-based system, decision or regression tree	High transparency
Work toward a goal across many tools	Agentic AI	Plans and acts, not just answers

Table A12. Benchmark companies mapped to course concepts

Company	What it shows	Course concepts
Delhivery (India)	NetMap address engine; dynamic mesh network routing; agentic real-time decisions	NLP, Transformers, knowledge representation, simulated annealing for VRP, agentic AI
Amazon Logistics (global)	Warehouse picking paths, layout optimisation, delivery routing, robotic pickers	A*, simulated annealing, genetic algorithms, reinforcement learning, goal-directed agents
Maersk (global)	Network digital twin, fuel-aware voyage planning, reefer container monitoring	Digital twin, heuristic search, green AI, industrial IoT and Edge AI predictive maintenance

Table A13. Stakeholder view (Unit 5 toolkit)

Stakeholder	Main concern	What the solution offers
Board / CEO	ROI and risk	OTIF and SLA protection; gated investment tied to pilot results

Stakeholder	Main concern	What the solution offers
CFO	Payback and budget control	Fuel and safety-stock savings measured against baselines; phased spend
CTO / IT	Technical feasibility and integration	Cloud and Edge integration with ERP, WMS and TMS
Middle management	Workload and relevance	Digital twin and demand dashboards that extend, not replace, tacit knowledge
Frontline staff	Job security and ease of use	Human-AI collaboration: fewer walking miles, safer driving, reskilling

Table A14. Curriculum alignment summary

Solution component	Course unit or framework	Concepts used
Dynamic route optimisation (CVRPTW, backhaul)	Unit 2: search and optimisation; last-mile framework	State space, A*, greedy, SA, GA, ACO, Tabu, reinforcement learning
NLP address geocoding	Unit 3: unstructured data; Unit 4: deep learning	NLP, Transformers, knowledge representation
IoT telematics and eco-driving	Unit 5: emerging trends; Scenario 6	Perception, Edge AI, industrial IoT, green AI
Predictive demand forecasting	Unit 4: ML and DL; Scenario 2	Limited memory AI, regression, trees and forests, LSTM, demand sensing, digital twin
Smart warehouse	Units 2, 4 and 5	A*, ACO, RL, multi-agent systems, CNNs, association rules, SA layout, anomaly detection
Roadmap and KPI matrix	Unit 5: implementation toolkit	Three horizons, readiness matrix, stakeholder groups, human-in-the-loop change

Table A15. How the two source reports were combined

Topic	Case report	Curriculum-mapping note	Treatment in this report
Core argument and structure	Strong: four-layer stack, worked example	Pillar structure, little argument	Case report structure kept
Technical depth	Core methods, A* worked example	Adds NLP, Edge AI, AMRs, predictive maintenance, ensembles	All merged into Modules 1 to 3
Benchmarks	Not covered	Delhivery, Amazon, Maersk	New Section 7 and Table A12
Roadmap	5 phases over 18 months with gates	3 horizons over 36 months	Phases nested inside horizons (Figure 3)
Numeric claims	None; stresses sourced savings only	OTIF, fuel and stock targets, unsourced	Shown only as planning targets (Figure 5)
Stakeholders and readiness	Governance and decision gates	Five stakeholders, 3x3 readiness matrix, Drishti case	Both kept (Sections 9, 10; Table A13)

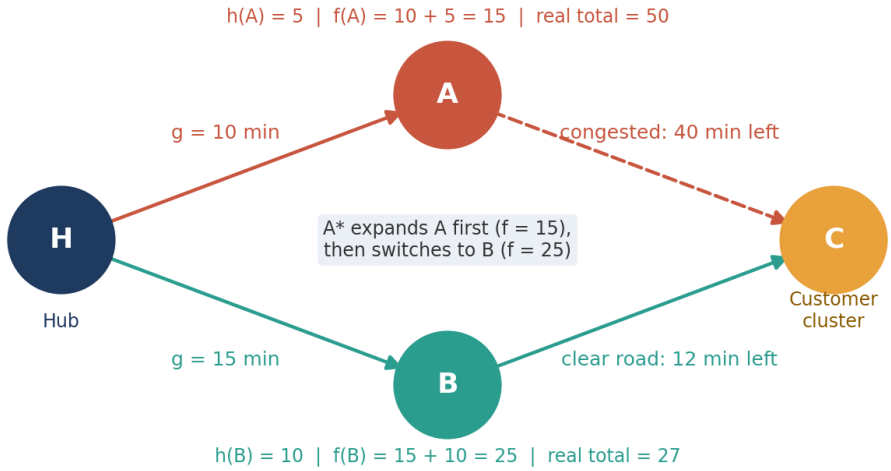
Figures

Figure 1. Route time: A* versus greedy



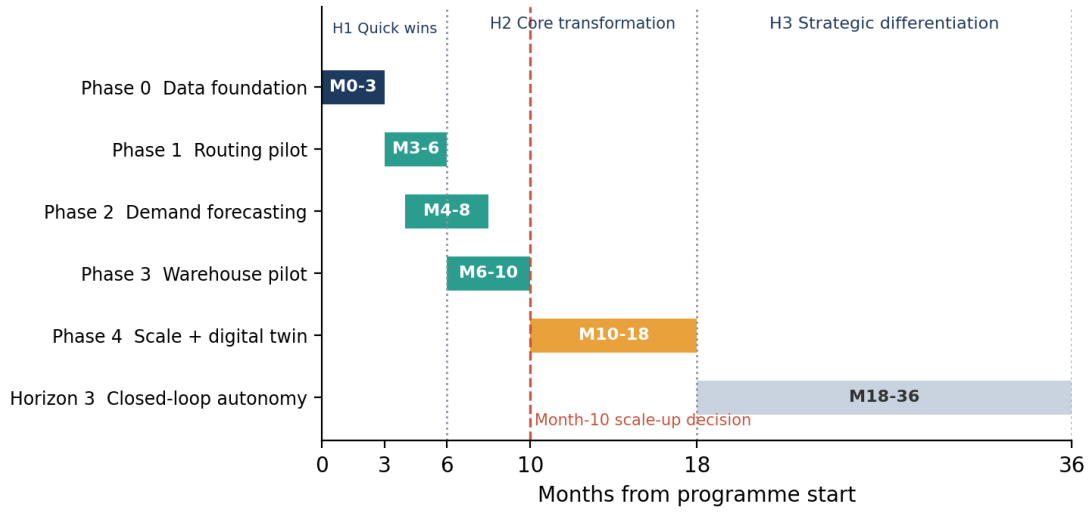
Greedy follows the smallest heuristic and is trapped behind congestion; A accounts for the real cost of each leg. Illustrative numbers from Table A4.*

Figure 2. Why A* switches from stop A to stop B



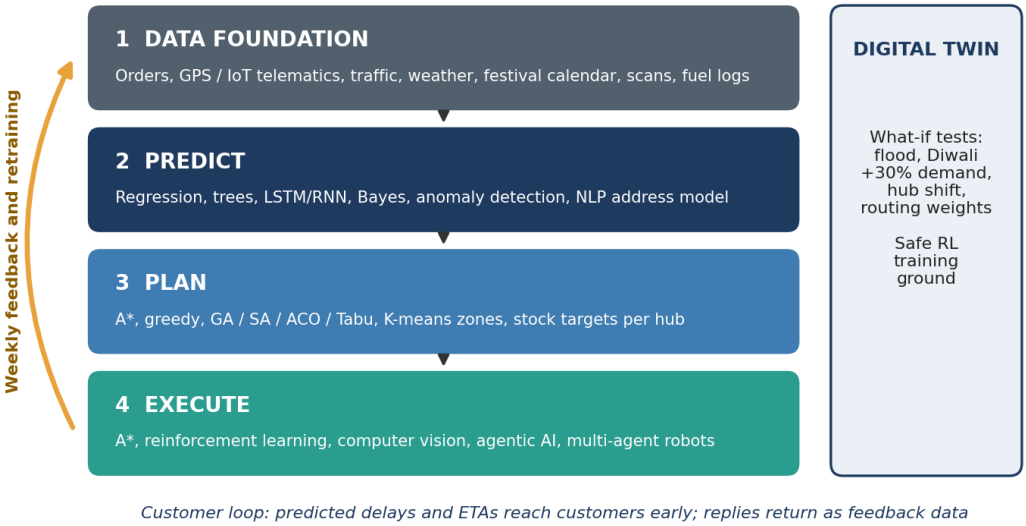
Stop A's estimate (f = 15) is lower than stop B's (f = 25), so A expands it first, but its true total of 50 minutes sends the search to stop B.*

Figure 3. Implementation roadmap and three-horizon view



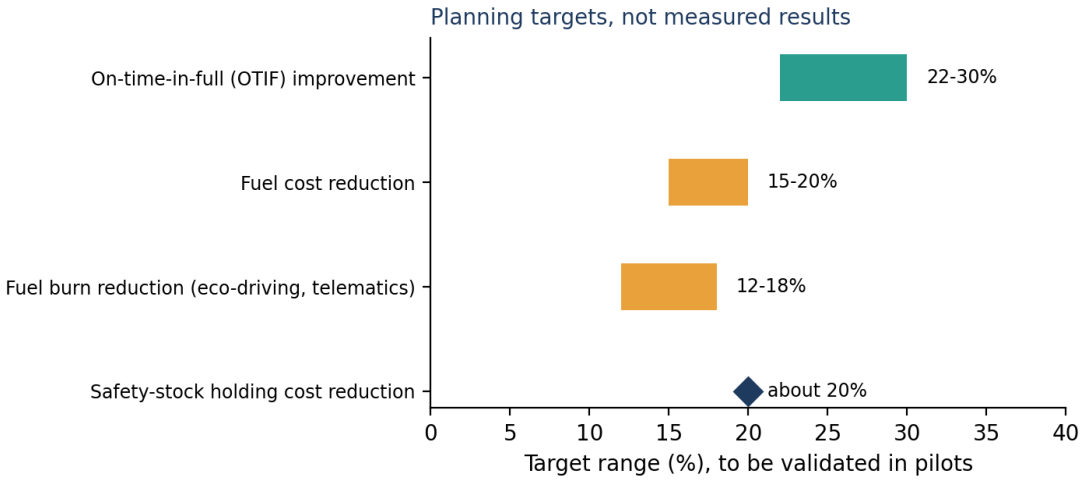
The three pilots overlap so routing, forecasting and warehouse results arrive in time for the month-10 scale-up decision.

Figure 4. Four-layer AI stack with feedback loop and digital twin



Each layer feeds the next; outcomes return weekly to retrain the Predict layer, and the digital twin tests decisions safely.

Figure 5. Indicative KPI targets



Ranges come from the curriculum-mapping draft and have no cited source. They are planning assumptions to be validated in pilots, not results.